

BASIC CHEMISTRY

Module map

- Module 02: Basic Chemistry
- Connected lab: lab-02_probability-statistics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Atoms and
elements

Bonding and
molecular
shape

Chemical
reactions in
cells

Module 02

Water as a
biological
solvent

Energy
and
chemical
balance

lab-02_probability-
statistics.md

BASIC CHEMISTRY

Learning objectives

- Use atomic structure to explain basic bonding patterns.
- Distinguish covalent, ionic, and hydrogen bonding in biological examples.
- Explain why water properties matter for cells and organisms.
- Interpret pH as a chemical condition that affects biology.
- Track atoms through a simple chemical reaction.

OBJECTIVE 1

Use atomic structure to explain basic bonding patterns.

OBJECTIVE 2

Distinguish covalent, ionic, and hydrogen bonding in biological examples.

OBJECTIVE 3

Explain why water properties matter for cells and organisms.

OBJECTIVE 4

Interpret pH as a chemical condition that affects biology.

OBJECTIVE 5

Track atoms through a simple chemical reaction.

BASIC CHEMISTRY

Topic sequence

- Atoms and elements: Atoms form elements, and their electrons shape bonding behavior.
- Bonding and molecular shape: Covalent, ionic, and hydrogen bonds create different molecular properties.
- Water as a biological solvent: Water polarity explains cohesion, temperature buffering, and dissolving ions or polar molecules.
- pH and chemical balance: pH tracks hydrogen ion concentration and affects biological molecules.
- Chemical reactions in cells: Cells depend on chemical reactions that rearrange matter without destroying atoms.

TOPIC 1

Atoms and elements

TOPIC 2

Bonding and molecular shape

TOPIC 3

Water as a biological solvent

TOPIC 4

pH and chemical balance

TOPIC 5

Chemical reactions in cells

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Module 02: Basic Chemistry Evidence Map

- Module focus: Basic Chemistry
- Central claim: Basic Chemistry explains atomic structure with water, pH, and biological molecule behavior.
- Key nodes to connect: Atoms and elements, Bonding and molecular shape, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-02_probability-statistics.md supplies an evidence surface for the map.

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Module 02: Basic Chemistry Reasoning Sequence

- Module focus: Basic Chemistry
- Trace the model: Atoms and elements -> Bonding and molecular shape -> Water as a biological solvent -> pH and chemical balance
- Inputs become outputs: Atoms and elements, Bonding and molecular shape -> Use atomic structure to explain basic bonding patterns., Distinguish covalent, ionic, and hydrogen bonding in biological examples.
- Feedback check: lab-02_probability-statistics.md evidence checks whether students can explain water as a biological solvent.
- Lab connection: lab-02_probability-statistics.md gives students a process to observe or test.

BASIC CHEMISTRY

Terms as evidence handles

- Atom: Smallest unit of an element that keeps that element identity.
- Element: Pure substance defined by proton number.
- Covalent bond: Bond formed when atoms share electrons.
- Ionic bond: Attraction between oppositely charged ions.
- Hydrogen bond: Weak attraction involving polar hydrogen and electronegative atoms.
- Polarity: Uneven charge distribution across a molecule.

ATOM

Smallest unit of an element that keeps that element identity.

ELEMENT

Pure substance defined by proton number.

COVALENT BOND

Bond formed when atoms share electrons.

IONIC BOND

Attraction between oppositely charged ions.

HYDROGEN BOND

Weak attraction involving polar hydrogen and electronegative atoms.

POLARITY

Uneven charge distribution across a molecule.

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Lab connection

- Lab file: lab-02_probability-statistics.md
- Lab evidence should test or illustrate: Water as a biological solvent
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION

Atoms and elements

EVIDENCE

lab-02_probability-statistics.md

CLAIM

Use atomic structure to explain basic bonding patterns.

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Contrast check

- Do not stop at: Atoms form elements, and their electrons shape bonding behavior.
- Stronger explanation: Cells depend on chemical reactions that rearrange matter without destroying atoms.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Atoms form elements, and their electrons shape bonding behavior.

DEEPER

Cells depend on chemical reactions that rearrange matter without destroying atoms.

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Module 02: Basic Chemistry Retrieval and Lab Check

- Module focus: Basic Chemistry
- Retrieval prompt: How do protons, neutrons, and electrons differ?
- Answer check: Use atomic structure to explain basic bonding patterns.
- Required terms: Atom, Element, Covalent bond
- Lab connection: lab-02_probability-statistics.md; lab-02_probability-statistics.md supplies evidence for probability and statistics for chemical evidence claims.

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Practice quiz bridge

- 1. Which statement best defines Atom in Module 02?
- 2. Which learning goal best supports the topic "Bonding and molecular shape"?
- 3. How should lab-02_probability-statistics.md support Module 02?
- 4. Which retrieval move best prepares a student for Basic Chemistry?

QUIZ 1

A

QUIZ 2

B

QUIZ 3

C

QUIZ 4

D

BASIC CHEMISTRY

Synthesis and exit ticket

- Exit claim: explain atoms and elements using evidence from lab-02_probability-statistics.md.
- Must include: Atom, Element, Covalent bond.
- Revision prompt: what evidence would change your explanation?

CLAIM

Atoms and elements

EVIDENCE

lab-02_probability-statistics.md

REVISION

What would change your mind?